THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	$\sqrt{x^2+y^2}$
2	**(b)** $(4,\ 6)$	Average the coordinates
3	**(c)** first quadrant	Signs of coordinates
4	**(a)** $(3,\ 0)$	$y=0$ on the x-axis
5	**(d)** 10	Distance formula
6	**(b)** 5	Abscissa is the x-coordinate
7	**(c)** 13	$\sqrt{x^2+y^2}$
8	**(b)** $(4,\ 3)$	Average the coordinates
9	**(b)** second quadrant	Signs of coordinates
10	**(c)** $(-4,\ 0)$	$y=0$ on the x-axis

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $=\sqrt{(3)^2+(4)^2}=\sqrt{25}$ (1); $=5$ units (1).	1+1+1
12	Midpoint $=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$ (1); $=(5,\ 8)$ (1).	1+1
13	Quadrant I (1); reflection $(3,-5)$ (1).	1+1
14	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $=\sqrt{(3)^2+(4)^2}=\sqrt{25}$ (1); $=5$ units (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $=\sqrt{(3)^2+(4)^2}=\sqrt{25}$ (1); $=5$ units (1).	1+1+1
16	Midpoint $=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$ (1); $=(5,\ 5)$ (1).	1+1
17	Section formula $\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$ (1); $x=\frac{1(6)+2(0)}{3}=2$ (1); $y=3$ (1).	1+1+1
	OR: $d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $=\sqrt{(8)^2+(6)^2}=\sqrt{100}$ (1); $=10$ units (1).	1+1+1
18	Midpoint $=\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$ (1); $=(-1,\ 3)$ (1).	1+1
	OR: Section formula $\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$ (1); $x=\frac{1(8)+1(2)}{2}=5$ (1); $y=4$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $=\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$ (1); $=\frac{1}{2}\lvert 0+12+0\rvert$ (1); $=6$ sq. units (1).	1+1+1
	OR: Centroid $=\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$ (1); $=(2,\ 3)$ (2).	1+2
20	Area $=\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$ (1); $=\frac{1}{2}\lvert -5+25+0\rvert$ (1); $=10$ sq. units (1).	1+1+1
	OR: Centroid $=\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$ (1); $=(4,\ 4)$ (2).	1+2

Mistakes to avoid:

- Using (x_1-x_2) and (y_2-y_1) inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 13	Distance formula
2	**(b)** 7	Abscissa is the x-coordinate
3	**(b)** 17	$\sqrt{x^2+y^2}$
4	**(c)** $(1,\ 5)$	Average the coordinates
5	**(c)** third quadrant	Signs of coordinates
6	**(c)** $(5,\ 0)$	$y=0$ on the x-axis
7	**(d)** 15	Distance formula
8	**(a)** -4	Abscissa is the x-coordinate
9	**(d)** 10	$\sqrt{x^2+y^2}$
10	**(b)** $(5,\ 3)$	Average the coordinates

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Midpoint $=\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $=(1,\ 1)$ (1).	1+1
12	Quadrant II (1); reflection $(-4, -2)$ (1).	1+1
13	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $=\sqrt{(3)^2+(4)^2}=\sqrt{25}$ (1); $=5$ units (1).	1+1+1
14	Midpoint $=\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $=(5,\ 10)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $=\sqrt{(5)^2+(12)^2}=\sqrt{169}$ (1); $=13$ units (1).	1+1+1
16	Midpoint $=\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $=(7,\ 1)$ (1).	1+1
17	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x=\frac{2(11)+3(1)}{5}=5$ (1); $y=5$ (1).	1+1+1
	OR: $d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ (1); $=\sqrt{(3)^2+(4)^2}=\sqrt{25}$ (1); $=5$ units (1).	1+1+1
18	Midpoint $=\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $=(-3,\ 5)$ (1).	1+1
	OR: Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x=\frac{3(10)+2(0)}{5}=6$ (1); $y=10$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $=\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$ (1); $=\frac{1}{2}\lvert -10+30+0\rvert$ (1); $=10$ sq. units (1).	1+1+1
	OR: Centroid $=\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $=(2,\ 2)$ (2).	1+2
20	Area $=\frac{1}{2}\lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\rvert$ (1); $=\frac{1}{2}\lvert 0+30+0\rvert$ (1); $=15$ sq. units (1).	1+1+1
	OR: Centroid $=\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $=(5,\ 5)$ (2).	1+2

Mistakes to avoid:

- Using (x_1-x_2) and (y_2-y_1) inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** fourth quadrant	Signs of coordinates
2	**(d)** $(-7,\ 0)$	$y = 0$ on the x-axis
3	**(c)** 17	Distance formula
4	**(b)** 6	Abscissa is the x-coordinate
5	**(c)** 29	$\sqrt{x^2 + y^2}$
6	**(d)** $(-1,\ 2)$	Average the coordinates
7	**(a)** first quadrant	Signs of coordinates
8	**(a)** $(6,\ 0)$	$y = 0$ on the x-axis
9	**(a)** 25	Distance formula
10	**(d)** 8	Abscissa is the x-coordinate

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Quadrant III (1); reflection $(-6, 3)$ (1).	1+1
12	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(8)^2 + (6)^2} = \sqrt{100}$ (1); $= 10$ units (1).	1+1+1
13	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (2,\ 1)$ (1).	1+1
14	Quadrant IV (1); reflection $(5, 4)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(6)^2 + (8)^2} = \sqrt{100}$ (1); $= 10$ units (1).	1+1+1
16	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (7,\ 4)$ (1).	1+1
17	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{1(4)+2(-2)}{3} = 0$ (1); $y = 2$ (1).	1+1+1
	OR: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ (1); $= \sqrt{(8)^2 + (15)^2} = \sqrt{289}$ (1); $= 17$ units (1).	1+1+1
18	Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ (1); $= (2,\ 2)$ (1).	1+1
	OR: Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{3(9)+1(1)}{4} = 7$ (1); $y = 4$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $= \frac{1}{2}\lvert x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\rvert$ (1); $= \frac{1}{2}\lvert 6 + 18 + 0\rvert$ (1); $= 12$ sq. units (1).	1+1+1
	OR: Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (3,\ 5)$ (2).	1+2
20	Area $= \frac{1}{2}\lvert x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\rvert$ (1); $= \frac{1}{2}\lvert -5 + 25 + 0\rvert$ (1); $= 10$ sq. units (1).	1+1+1
	OR: Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (1,\ 4)$ (2).	1+2

Mistakes to avoid:

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